/Users/frankie/Dropbox/Harper College/CIS 2026/M12/M12P3.py

function P1.py | M11 More on function P2.py | M11 Advanced function P1.py | M11 Advanced function P2.py | M12P3.py

```python
lastNameList = []
examScoreList = []

# Function for printing the array
def print_array(arr, titleOfList):
    #Print the array
    print(titleOfList)
    print("="*20)
    for val in arr:
        print(val)

# Function for reverse print
def print_reverse_array(arr,titleOfList):
    # Reverse the array
    arr.reverse()
    print_array(arr,titleOfList)

# Function for parallel print
def print_parallel(arr1,arr2,titleOfList):
    print(titleOfList)
    print("="*35)
    for index in range(len(arr1)):
        print(f"{arr1[index].ljust(30)}{arr2[index]:5n}")

# Function for Reverse parallel print
def print_reverse_parallel(arr1,arr2,titleOfList):
    arr1.reverse()
    arr2.reverse()
    print_parallel(arr1, arr2, titleOfList)

# Function for loading data
def load_data():
    # Declare List 1 and list 2
    list1 = []
    list2 = []

    f = open(FILENAME, 'r')
    while True:
        tmp1 = f.readline().rstrip("\n")
        if not tmp1:
            break;
        # Append to list1
        list1.append(tmp1)
        tmp2 = float(f.readline().rstrip("\n"))

        # Append to Score List
        list2.append(tmp2)
    return list1, list2
    f.close()

# Compare and Find the Highest/Lowest scores
def compare_score(names,scores,isHighest = True):

    lastScoreIndex = []
    lastScoreIndex.append(0)

    # Set the flag = 1 if searching highest score else flag = -1
    flag = 1 if isHighest else -1

    # Compare the score with latest Highest/Lowest score
    for index in range(1,len(scores)):
        if scores[index] * flag > scores[lastScoreIndex[0]] * flag:

            # if new Highest/Lowest score find, replace the old list
            lastScoreIndex.clear()
            lastScoreIndex.append(index)

        elif scores[index] * flag == scores[lastScoreIndex[0]] * flag:

            # if more then one Highest/Lowest Scores, append into the list
            lastScoreIndex.append(index)


    # Display the result
    print(f"\n{"Highest" if isHighest else "Lowest"} Score")
    print(f"{"Last Name".ljust(30," ")}{"Score".rjust(5," ")}")
    print("="*35)
    for index in lastScoreIndex:
        print(f"{names[index].ljust(30)}{scores[index]:5n}")



def main():
    lastNameList,examScoreList = load_data()

    # Print in Normal order
    print_parallel(lastNameList,examScoreList,f"Normal order\n{"Last Name".ljust(30," ")}{"Score".rjust(5," ")}")

    # print reverse list of last name
    print_reverse_parallel(lastNameList,examScoreList, f"\nReverse order\n{"Last Name".ljust(30," ")}{"Score".rjust(5," ")}")

    # Find the Highest Score
    compare_score(lastNameList, examScoreList, True)

    # Find the Lowest Score
    compare_score(lastNameList, examScoreList, False)
```

Name	Date Modified
norton	3/19/26 7:28 PM
M12P1.pdf	4/9/26 1:10 AM
M12P1.py	4/9/26 1:09 AM
M12P2.pdf	4/9/26 1:11 AM
M12P2.py	4/9/26 1:08 AM
M12P3 Data.txt	4/8/26 10:34 PM
M12P3.py	4/9/26 1:16 AM
M12P4 Data.txt	4/8/26 10:47 PM
M12P4.py	4/9/26 1:02 AM
M12P5.py	4/9/26 1:02 AM
Old	4/8/26 8:09 PM

Debugger | Profiler | Plots | Files

Console...

```
In [119]: %runfile '/Users/frankie/Dropbox/Harper College/CIS 2026/M12/M12P3.py' --wdir
Normal order
Last Name                     Score
===================================
Chan                             90
Brothers                         80
Kennth                           70
Vara                             60
Blish                            50
Batt                             95
Cho                              85
Gend                             50
Tropp                            95
Herro                            55

Reverse order
Last Name                     Score
===================================
Herro                            55
Tropp                            95
Gend                             50
Cho                              85
Batt                             95
Blish                            50
Vara                             60
Kennth                           70
Brothers                         80
Chan                             90

Highest Score
Last Name                     Score
===================================
Tropp                            95
Batt                             95

Lowest Score
Last Name                     Score
===================================
Gend                             50
Blish                            50

In [120]:
```

IPython Console | History

Inline | Conda: spyder-runtime (Python 3.12.11) | LSP: Python | Line 86, Col 1 | UTF-8 | LF | RW | Mem 75%